

1. Executive Summary

In the analyzed dataset, 114 out of 400 customers (28.5%) were classified as Churn. This represents the proportion of customers classified as churned, not the monthly churn rate. This value depends on the assumption that the 286 missing values in the original `churn_status` column correspond to No Churn.

Differences in churn rates were observed across subscription plans, with substantially higher rates among Free plan subscribers. Simply having a recorded support ticket was not significantly associated with churn status. However, among customers with support tickets, those classified as Churn had longer average resolution times than those classified as No Churn. The level of recorded activity was associated with churn. Among customers with at least one recorded event, activities such as tracking and sharing workouts were associated with lower churn rates.

It is worth testing improvements in customer support efficiency and initiatives aimed at encouraging regular use of workout-related features, while simultaneously improving the collection of subscription cancellation data. The findings do not establish that the identified behaviors or customer experiences cause churn, as the datasets lack cancellation dates and may contain gaps in activity logs.

2. Business Problem and Analysis Objectives

Fit.ly is a subscription-based fitness application. The company is facing an increasing number of customer cancellations and is therefore exploring potential retention strategies, with particular emphasis on subscription plans, recorded user behavior, and interactions with customer support. Four key research questions were formulated:

- Does churn status differ across subscription plans?
- Is contacting customer support associated with churn, and does resolution time matter?
- How is the number of recorded activities associated with churn?
- Do the results differ by activity type?

The analysis is cross-sectional and exploratory rather than experimental. The provided dataset does not include cancellation dates or historical data on the number of active subscribers over consecutive months, which represents a significant limitation.

Data source:

- `da_fitly_account_info.csv`
- `da_fitly_customer_support.csv`
- `da_fitly_user_activity.csv`

3. Data Validation

3.1. Account info

customer_id

Validation result: 400 unique values, with no missing values or duplicates.

No changes.

email

Validation result: 400 non-null and unique values. Basic checks confirmed the presence of the @ symbol and the absence of spaces. Email addresses will not be used to draw conclusions about churn.

No changes.

state

Validation result: 50 U.S. state names; no missing values or invalid state names were found when checked against the validated list.

No changes.

plan

Validation result: Only Free, Basic, Pro, and Enterprise subscription plans were found, with no missing values.

No changes.

plan_list_price

Validation result: Numeric values range from 0 to 148, with no missing or negative values. The Free plan has a price of 0.

No changes.

churn_status

Validation result: 114 Y values and 286 missing values.

Missing values were replaced with N.

Signup_date

The column is not present in the dataset, despite being described by the Product Manager.

3.2. Customer support

ticket_time

Validation result: 918 valid dates ranging from June 5 to December 1, 2025, with no missing values.

Converted from object to datetime.

user_id

Validation result: 367 unique users (consistent with the fact that not all customers submitted support requests). No missing values.

No changes (The user_id will be standardized during data merging)

channel

Validation result: 4 categories: email (298), chat (294), phone (287), and "-" (hyphen) (39).

The "-" values were replaced with "Unknown".

topic

Validation result: 4 categories: billing: 239, other: 228, technical: 226, account: 225. No missing values.

No changes.

resolution_time_hours

Validation result: 918 values ranging from 0.52 to 32.46 hours. No missing or negative values.

No changes.

state

Validation result: Codes 0 and 1. Meaning of code was not explained in the documentation.

The column was renamed to "state_code" to avoid confusion when merging the data with account_info.

comments

Validation result: 46 comments and 872 missing values; the comments include requests for data deletion.

Missing values were left unchanged, as the comment content was not analyzed.

Duplicate support tickets were also checked based on the combination of user_id and ticket_time and no duplicates were detected. However, the absence of a separate ticket_id prevents full verification of the business-level uniqueness of all support tickets.

3.3. User activity

event_time

Validation result: 445 valid dates ranging from June 5 to December 1, 2025, with no missing values.

Converted to datetime.

user_id

Validation result: 246 unique users, with no missing values. This is consistent with the fact that the upstream system experiences downtime.

No changes (The user_id will be standardized during data merging)

event_type

Validation result: 4 categories: read_article: 125; watch_video: 120; track_workout: 108; share_workout: 92. No missing values.

No changes.

4. Data Preparation and Integration

User identifiers in the support and activity datasets were standardized to the customer_id format (C + user_id). The two detailed tables (support_full and activity_full) were then retained, and customer-level aggregations (one row per customer) were created (support_summary and activity_summary).

Only these aggregated tables were left-joined to the 400 customer accounts. This prevented row multiplication that would otherwise result from separately joining one-to-many relationships.

Missing values in count variables were replaced with 0 after merging, while resolution times for customers without support tickets were left as NaN.

Dataset	Description	Number of Rows
<code>support_full</code>	Individual support ticket enriched with customer account data	918
<code>support_summary</code>	Customer-level summary of support tickets	367
<code>activity_full</code>	Individual activity event enriched with customer account data	445
<code>activity_summary</code>	Customer-level summary of activity events	246
<code>customer_analysis</code>	Integrated dataset - one row per customer	400

Integration Validation Result: customer_analysis contains 400 rows and 22 columns. All 400 customer identifiers are unique, and the total numbers of activity events and support tickets match the source files: 445 and 918, respectively. Support tickets were submitted by 367 customers, while activity events were recorded for 246 customers.

5. Exploratory Data Analysis (EDA)

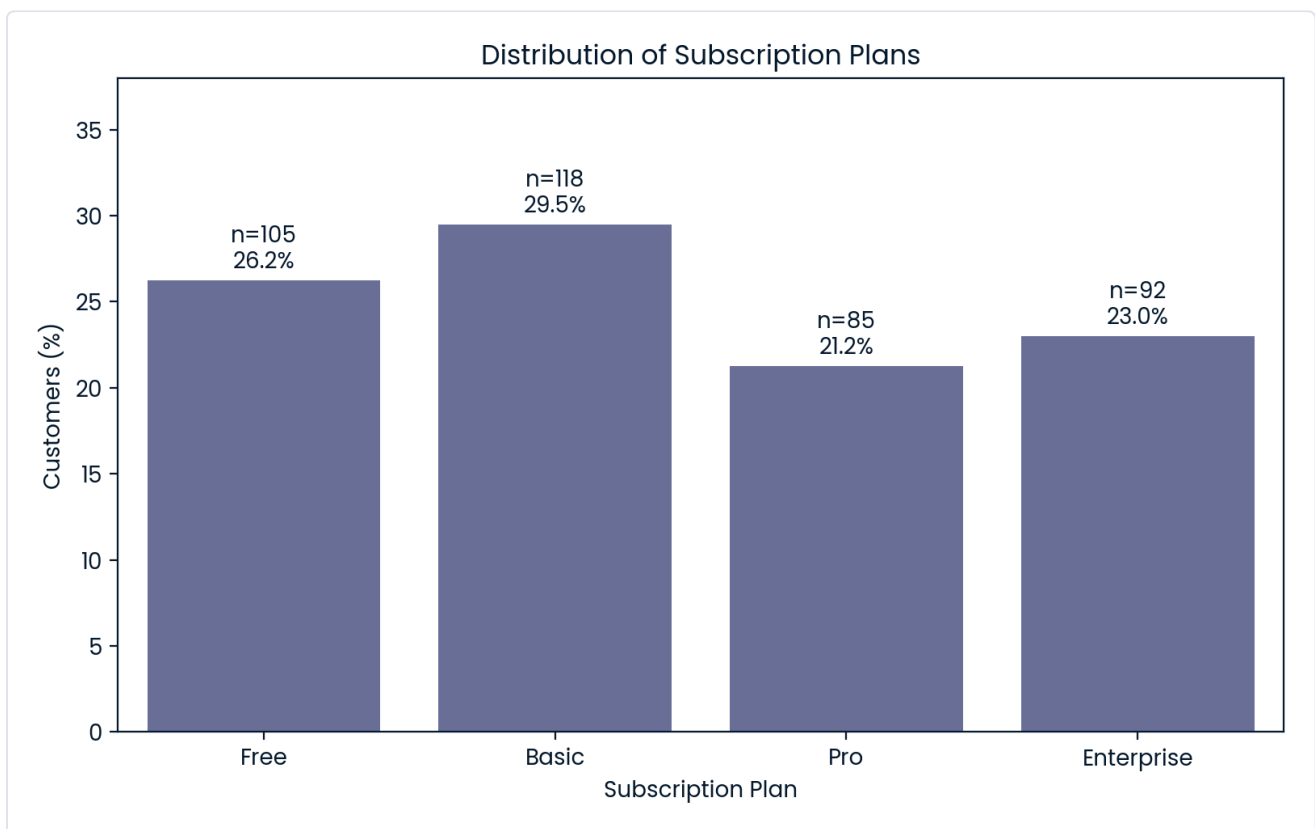
5.1. Univariate Analysis

Summary: The dataset includes four subscription plans, and 28.5% of accounts are classified as churned. No activity events were recorded for 38.5% of accounts. Support tickets are distributed relatively evenly across issue categories and known contact channels. None of these distributions, on their own, establishes the causes of churn.

Subscription Plan Distribution

How are subscription plans distributed among customers?

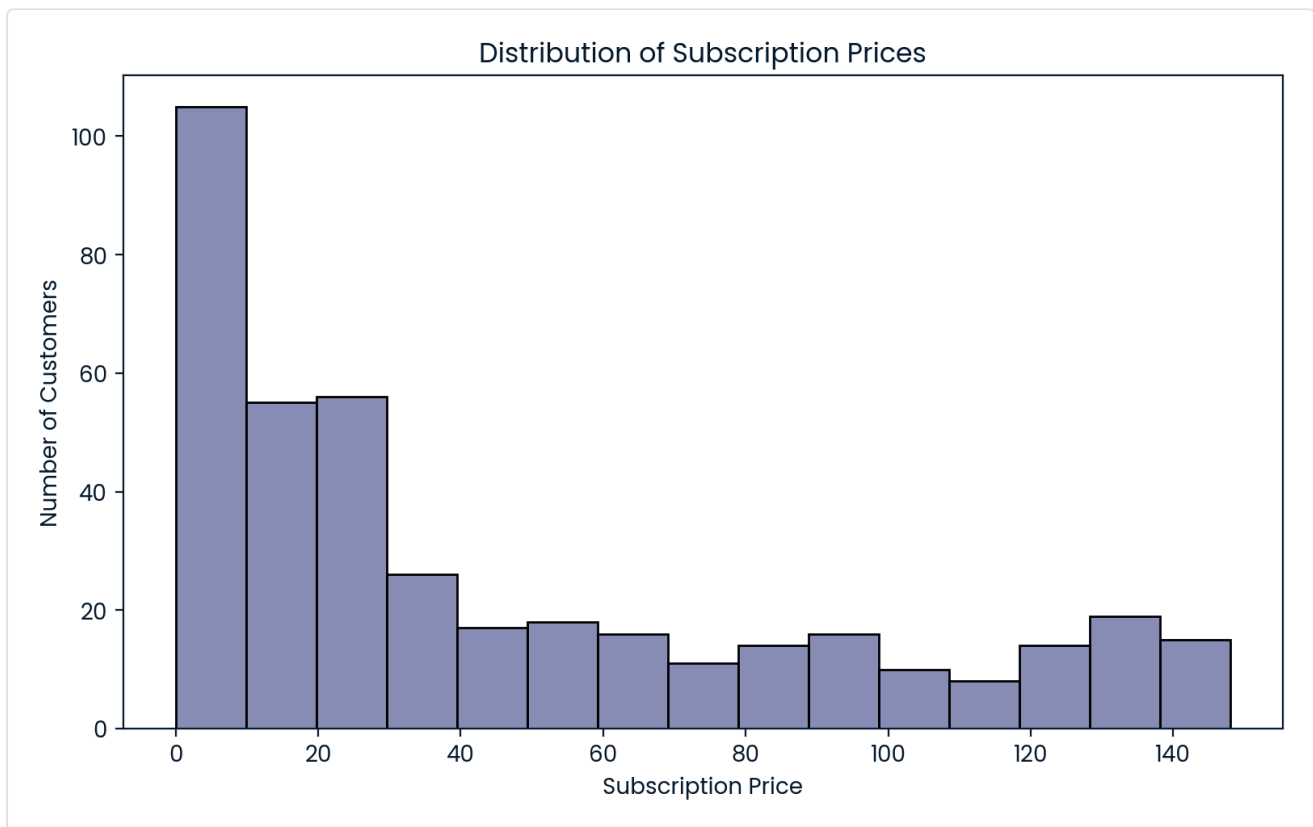
Basic - 118 customers (29.5%), Free - 105 (26.25%), Enterprise - 92 (23.0%), and Pro - 85 (21.25%). Each subscription plan represents a substantial share of the customer base.



Subscription Price

How are subscription prices distributed among customers?

Subscription prices range from 0 to 148, with a median of 26 and a mean of approximately 43.97. A price of 0 corresponds to the Free plan. The price distribution is associated with the subscription plan type and should not be interpreted as an independent effect on churn.

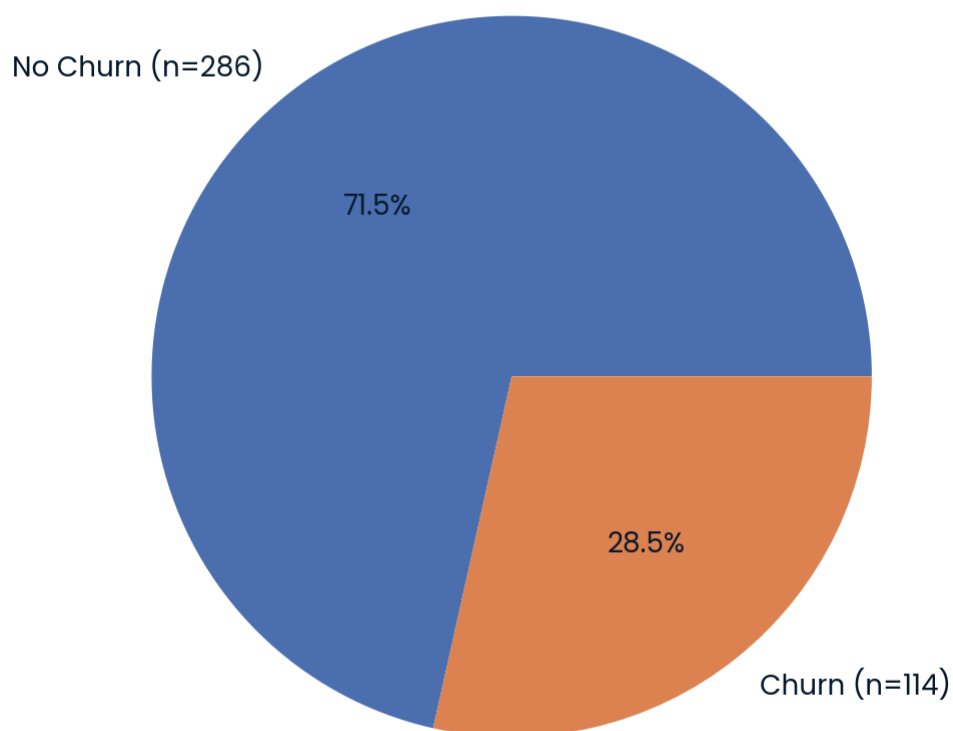


Churn Status

What proportion of the analyzed customer base is classified as churned?

Churn — 114 customers (28.5%); No Churn — 286 customers (71.5%).

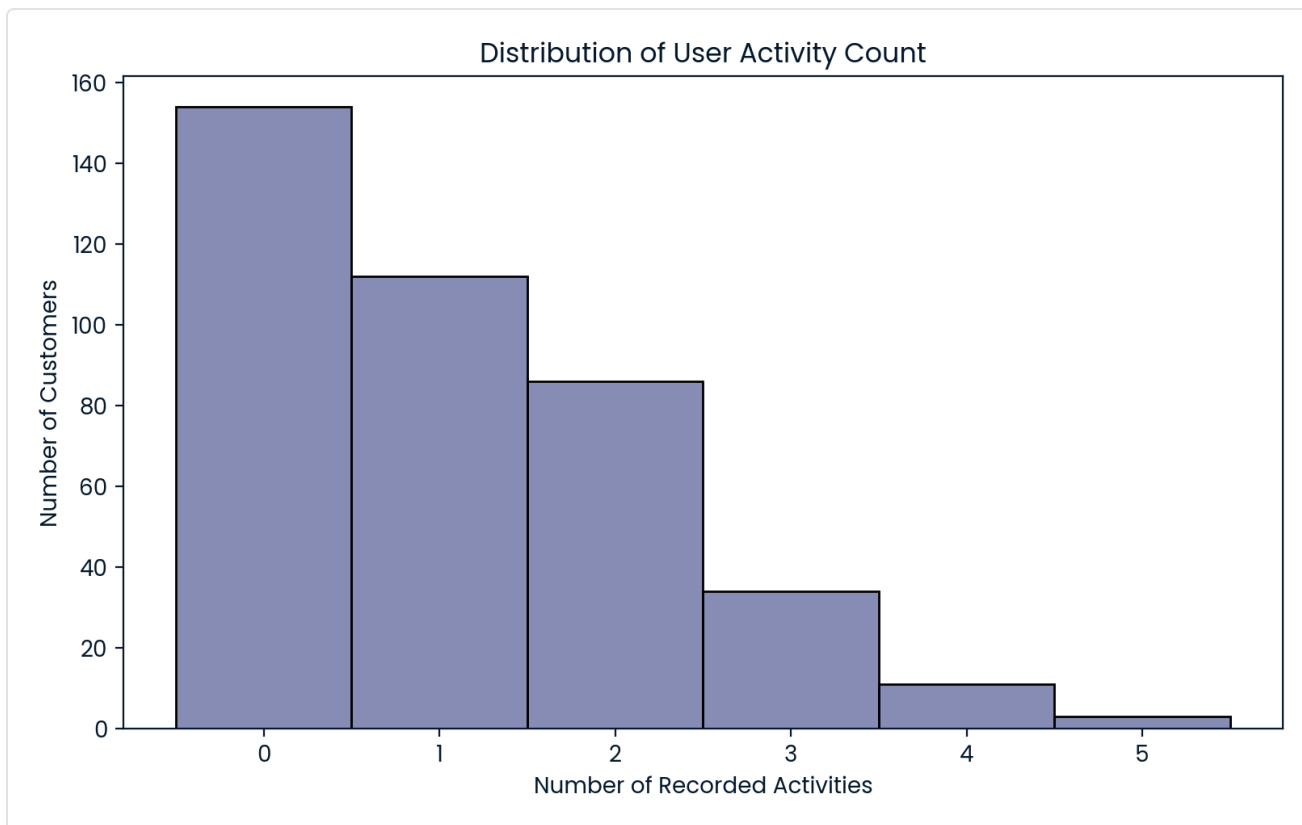
Distribution of Churn Status



Number of Activities per Customer

How many activity events were recorded for each customer?

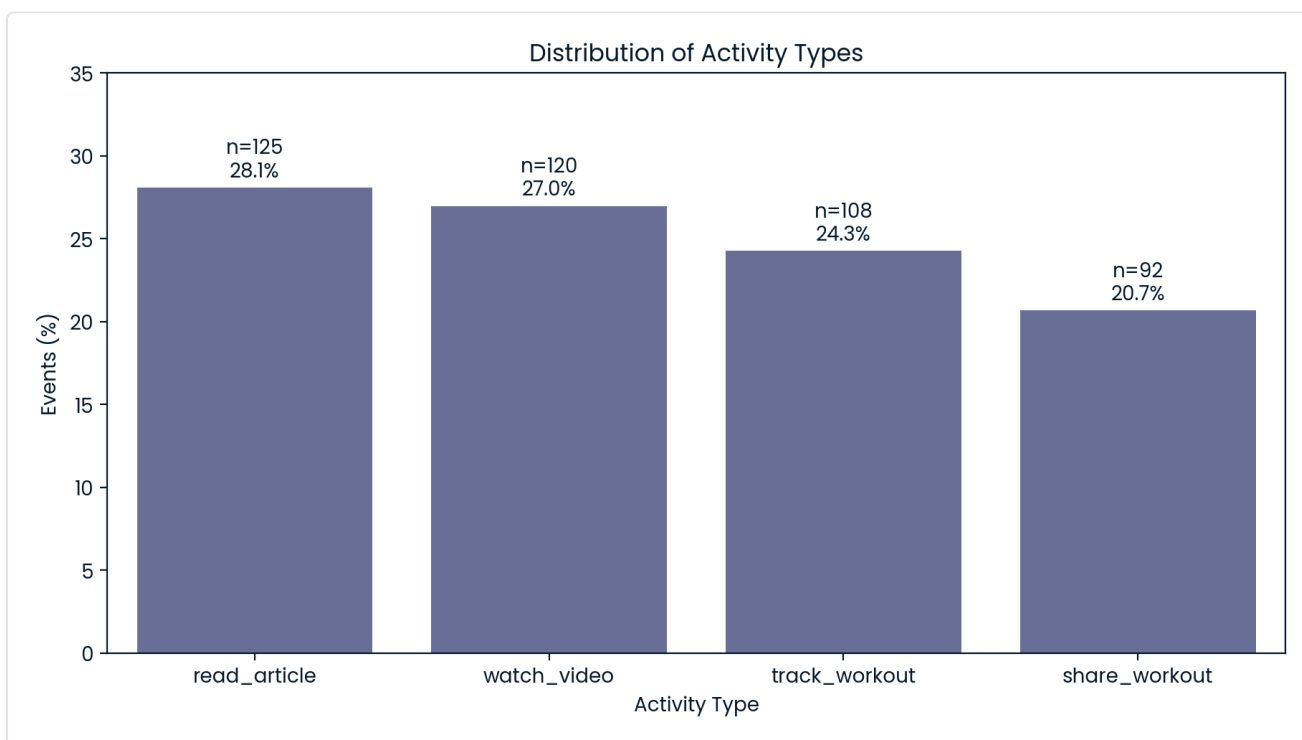
No activity events were recorded for 154 customers (38.5%). The mean number of events was approximately 1.11 per customer. A value of zero indicates the absence of recorded events in the provided dataset, rather than confirmed non-use of the application.



Types of Recorded Activities

Which activity events are recorded most frequently?

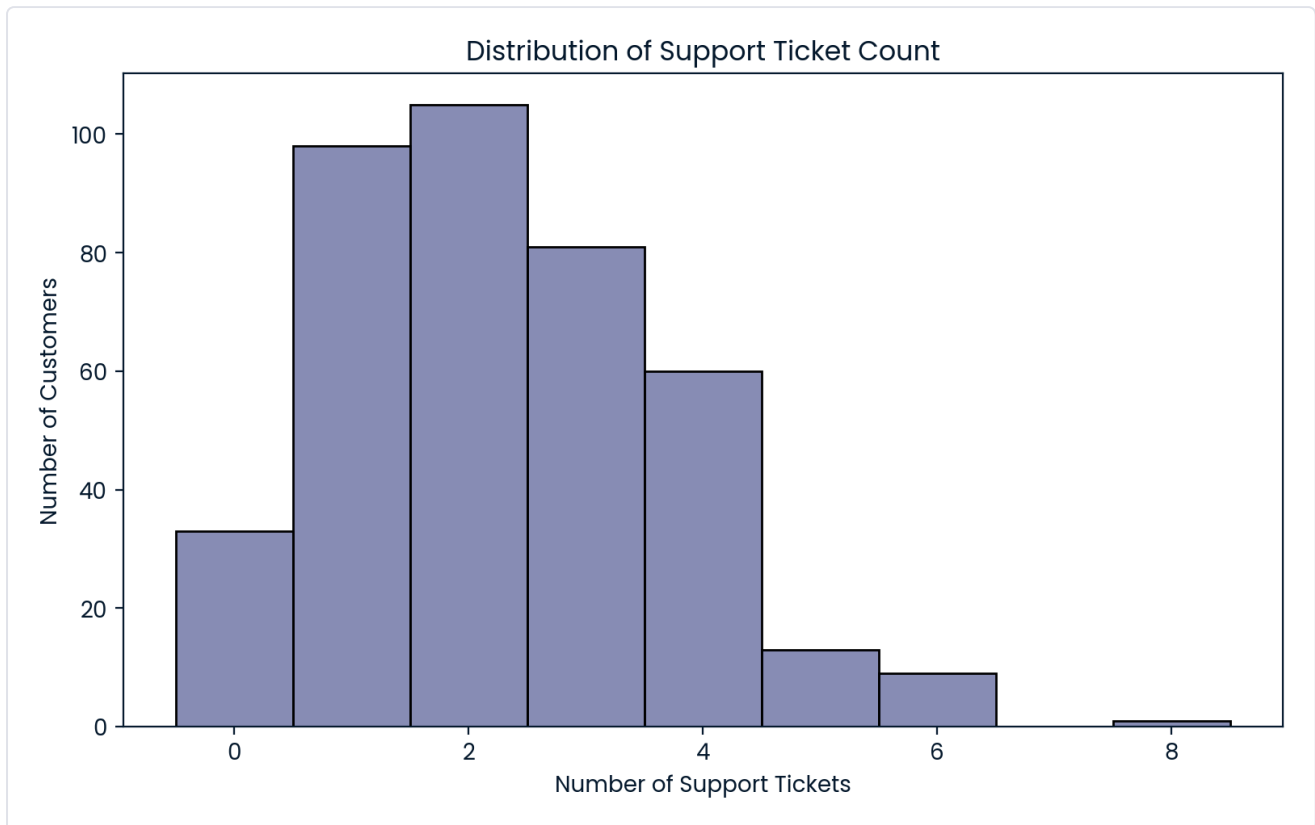
The most frequently recorded activities were `read_article` (125) and `watch_video` (120), followed by `track_workout` (108) and `share_workout` (92). Important: a single customer may account for multiple events.



Number of Support Tickets per Customer

How many support tickets were submitted per customer?

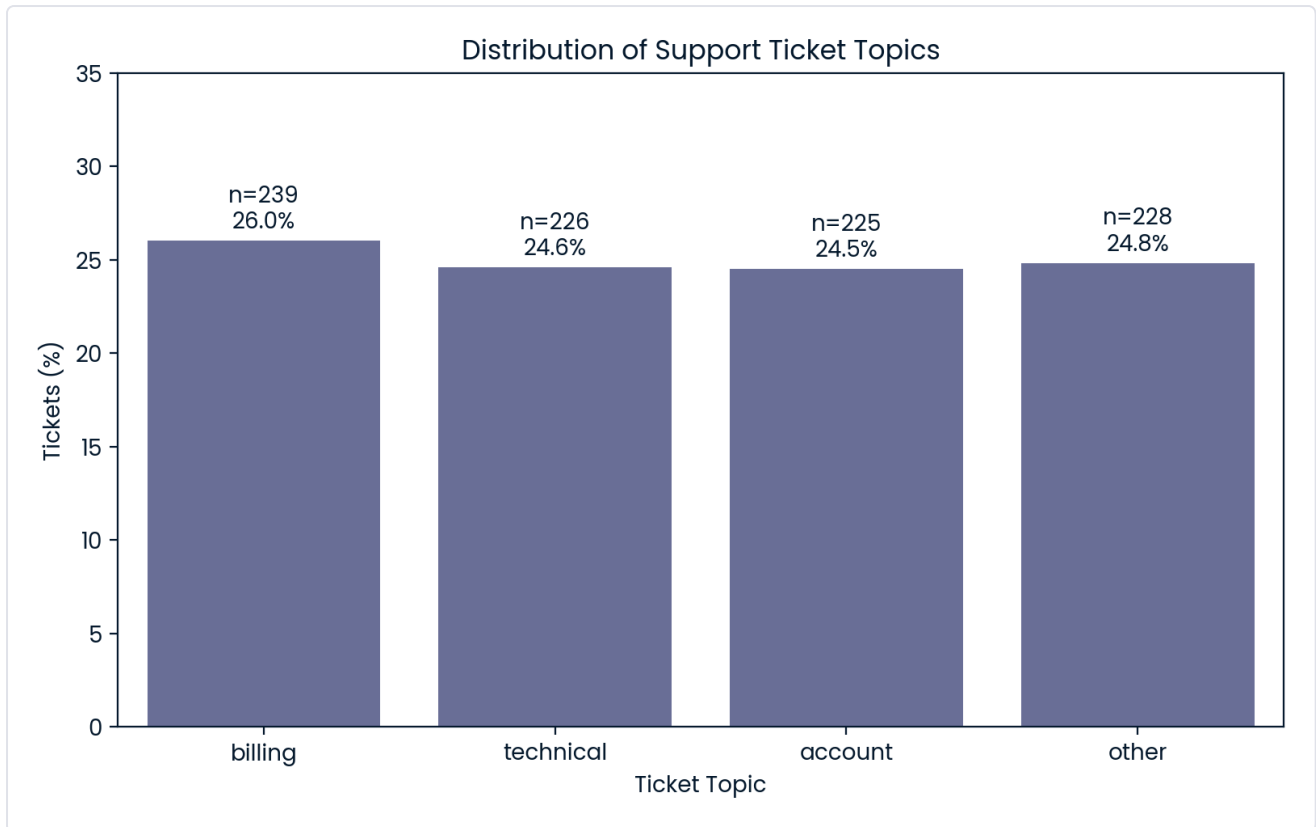
The 918 support tickets correspond to an average of 2.30 tickets per customer. No support tickets were recorded for 33 customers.



Support Ticket Categories

What types of issues do customers contact support about?

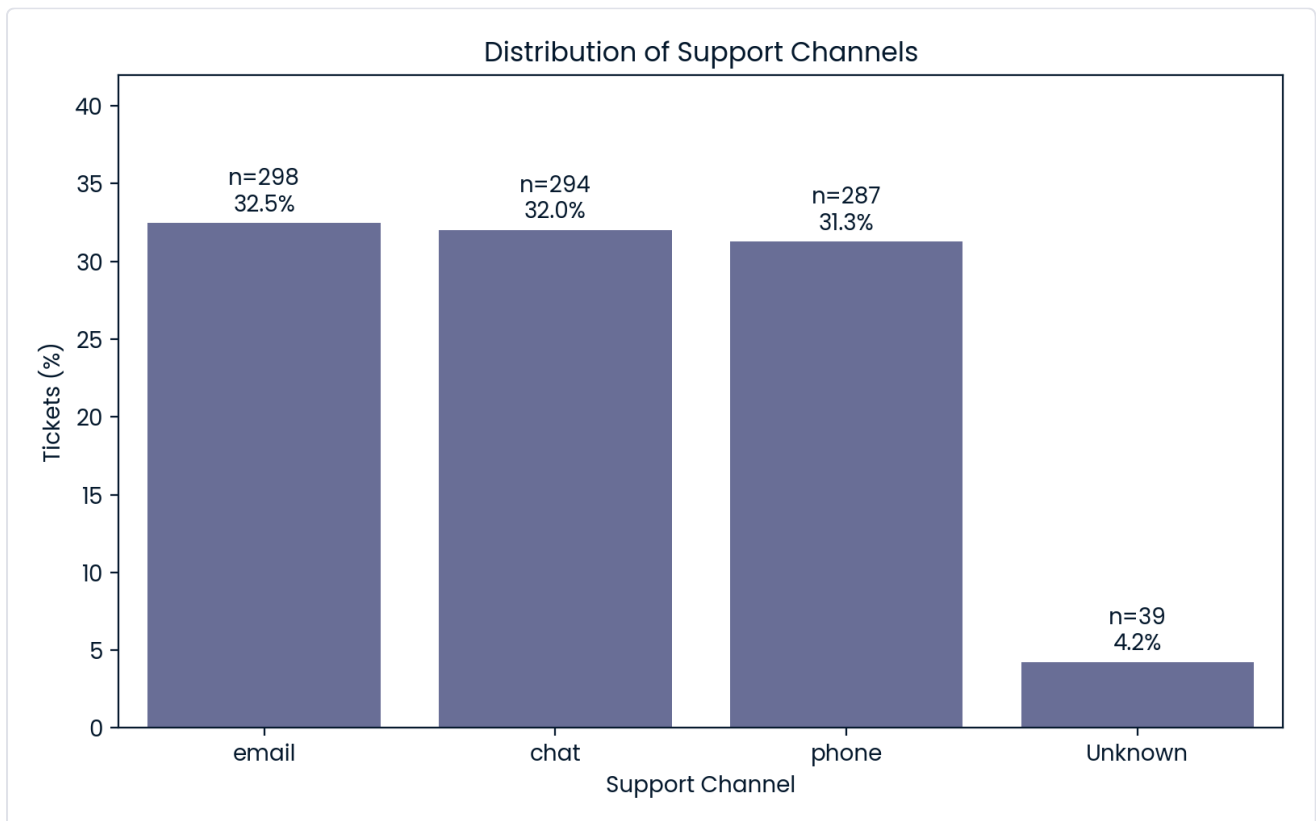
No single category clearly dominates the distribution of support tickets.



Support Contact Channels

Through which channels are support tickets submitted?

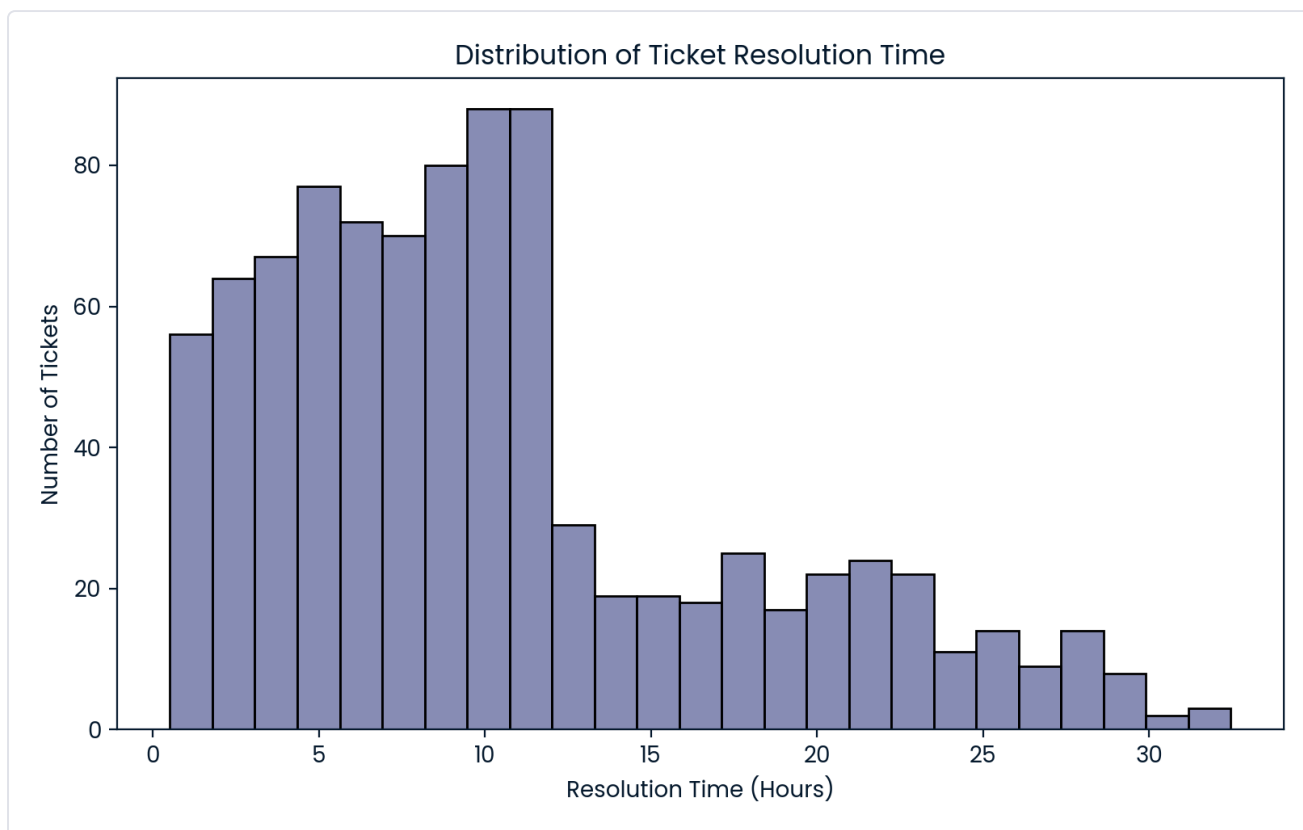
The three known contact channels have similar frequencies; Unknown indicates that the original contact channel information was missing.



Support Ticket Resolution Time

How are support ticket resolution times distributed?

Mean $\approx$ 10.39 hours, median $\approx$ 9.04 hours, range: 0.52–32.46 hours.



5.2. Multivariate Analysis

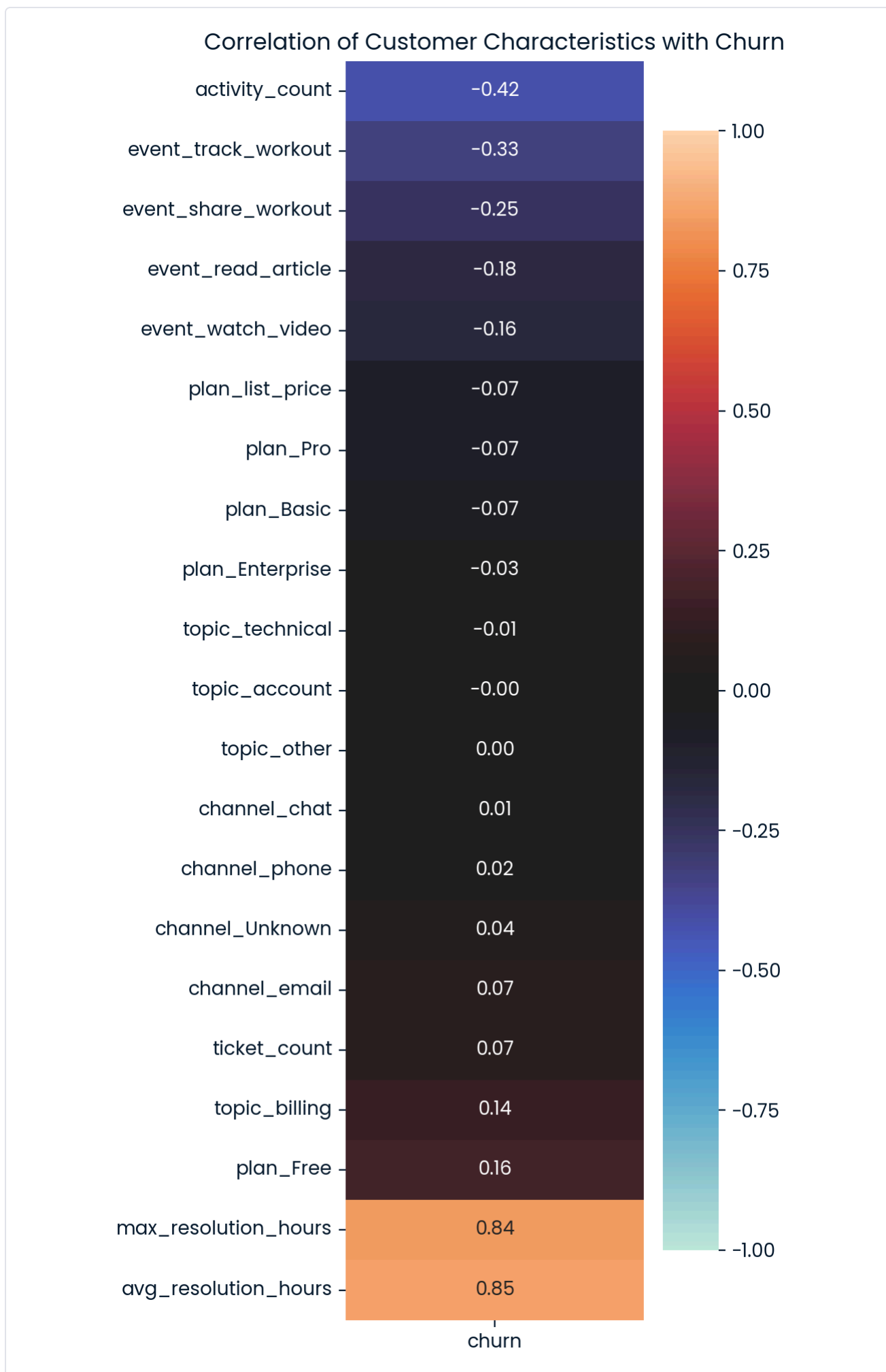
Method: Comparisons of churn proportions across groups and statistical hypothesis testing. Significance level: $\alpha = 0.05$. The tests examine associations between variables, not causal effects.

All analyses use `customer_analysis`. Where a variable is not applicable to a customer (e.g., resolution time for customers without support tickets), only the corresponding missing values are excluded.

Preliminary Analysis

Which variables are associated with churn status?

Churn status and subscription plans were encoded as binary variables (0/1), using one-hot encoding for subscription plans. Correlations were calculated pairwise: resolution time data were available for 367 customers, while most other variables had data for all 400 customers. The strongest observed associations were a positive correlation between average support ticket resolution time and churn (approximately 0.85) and a negative correlation between the number of recorded activities and churn (approximately -0.42). The correlation between Free plan membership and churn was positive but considerably weaker (approximately 0.17). The relationship between subscription plan and churn was analyzed separately.



Subscription Plan vs. Churn

Does the proportion of customers classified as churned differ across subscription plans?

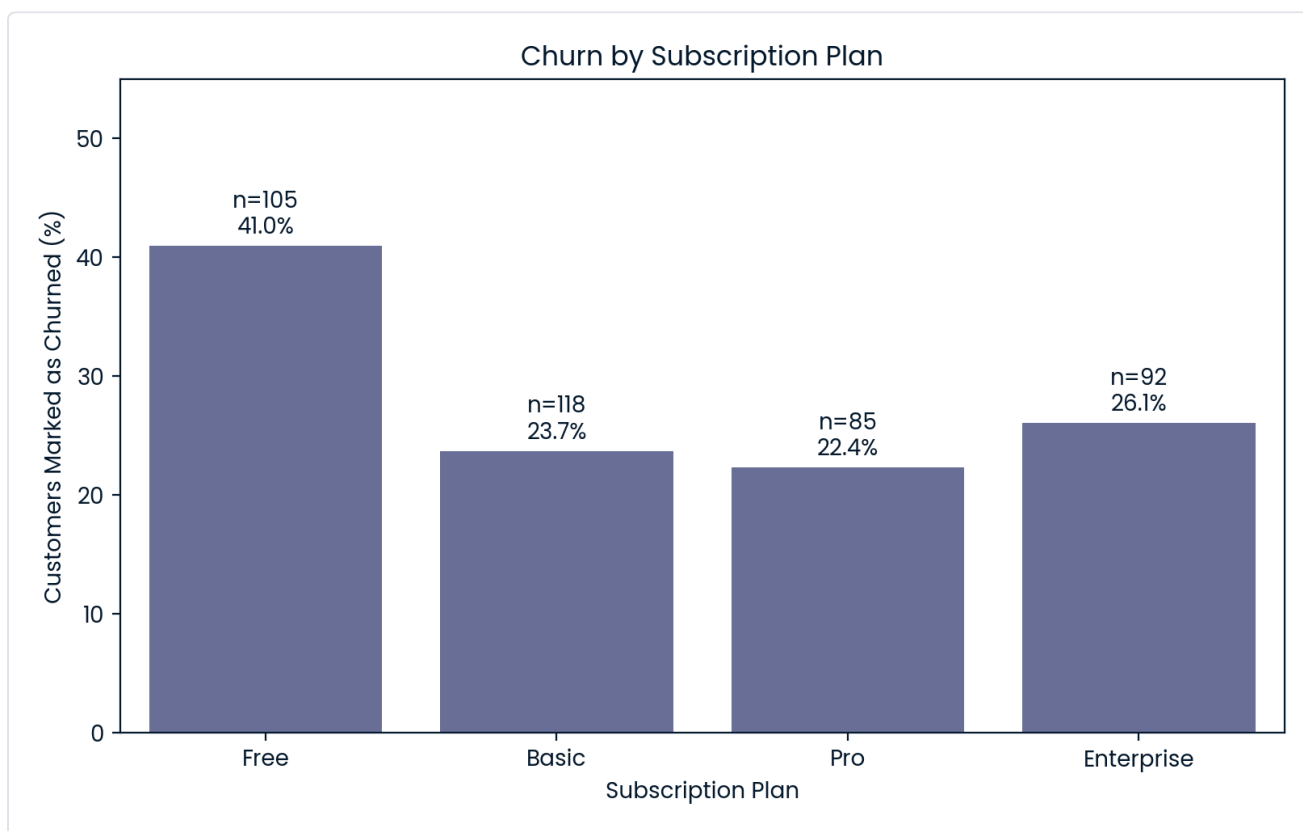
- H_0 : Subscription plan and churn are independent.
- H_1 : There is an association between subscription plan and churn.

The Free plan has the highest proportion of customers classified as churned (41.0%), followed by Enterprise (26.1%), Basic (23.7%), and Pro (22.4%).

The chi-square test ($\chi^2(3) = 11.15$, $p = 0.011$) indicates a statistically significant association in the dataset, although the effect size is small (Cramér's $V \approx 0.17$).

The data suggest that customers on the Free plan have the highest proportion of accounts classified as churned.

Plan	No Churn (N)	Churn (Y)	Total (n)	Churn (%)
Free	62	43	105	41.0
Basic	90	28	118	23.7
Pro	66	19	85	22.4
Enterprise	68	24	92	26.1



Support Contact vs. Churn

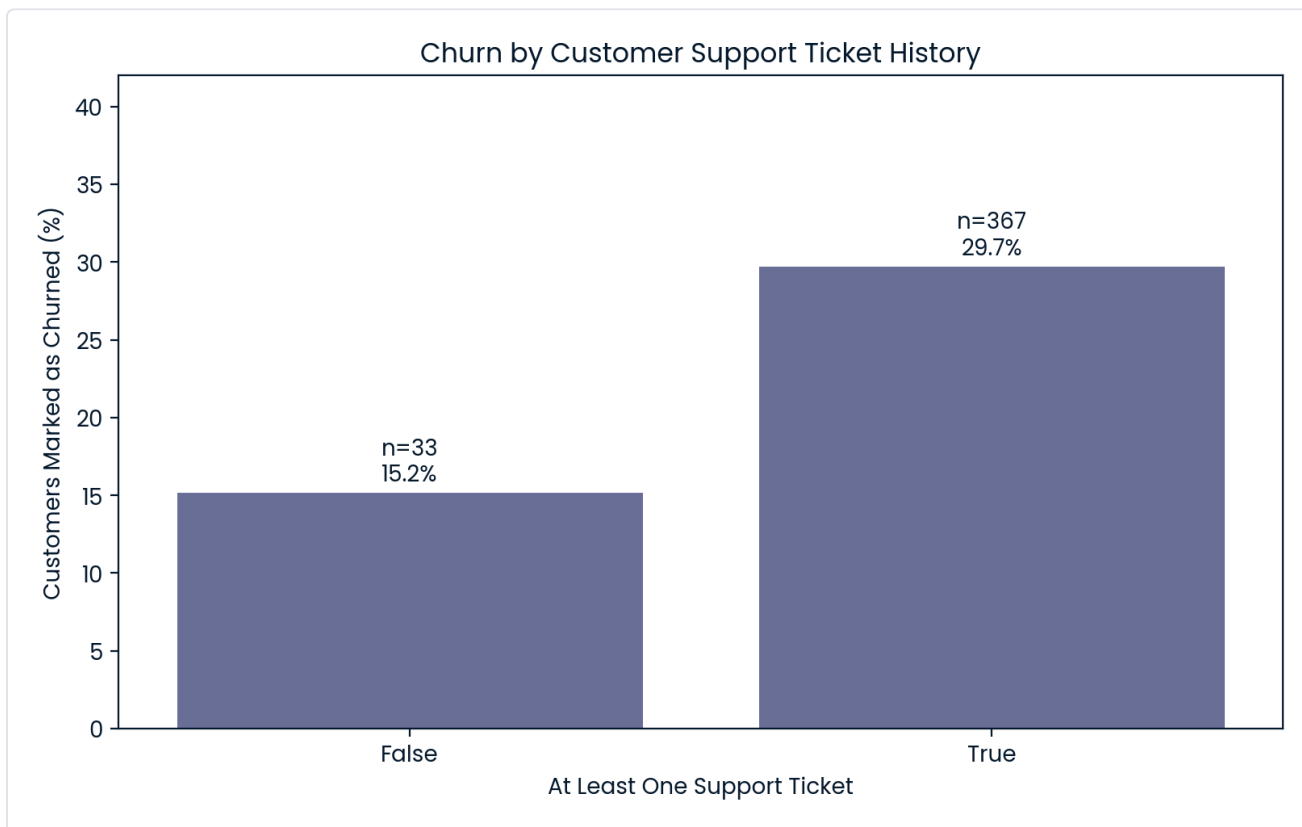
Is having at least one recorded support ticket associated with churn status?

- H_0 : Having at least one support ticket and churn status are independent.
- H_1 : There is an association between having at least one support ticket and churn status.

The churn rate among customers with at least one support ticket was 29.7% ($n = 367$), compared with 15.2% ($n = 33$) among customers without any support tickets. The difference was not statistically significant in the applied test ($\chi^2(1) = 2.47$, $p = 0.116$). No statistically significant association was established.

This does not constitute evidence that no association exists. The group of customers without support tickets was relatively small.

Has Ticket	No Churn (N)	Churn (Y)
False	28	5
True	258	109



Average Support Ticket Resolution Time vs. Churn

Do customers with churn status Y and N differ in their average support ticket resolution times?

- H_0 : The mean resolution times are equal across the two groups.
- H_1 : The mean resolution times differ between the two groups.

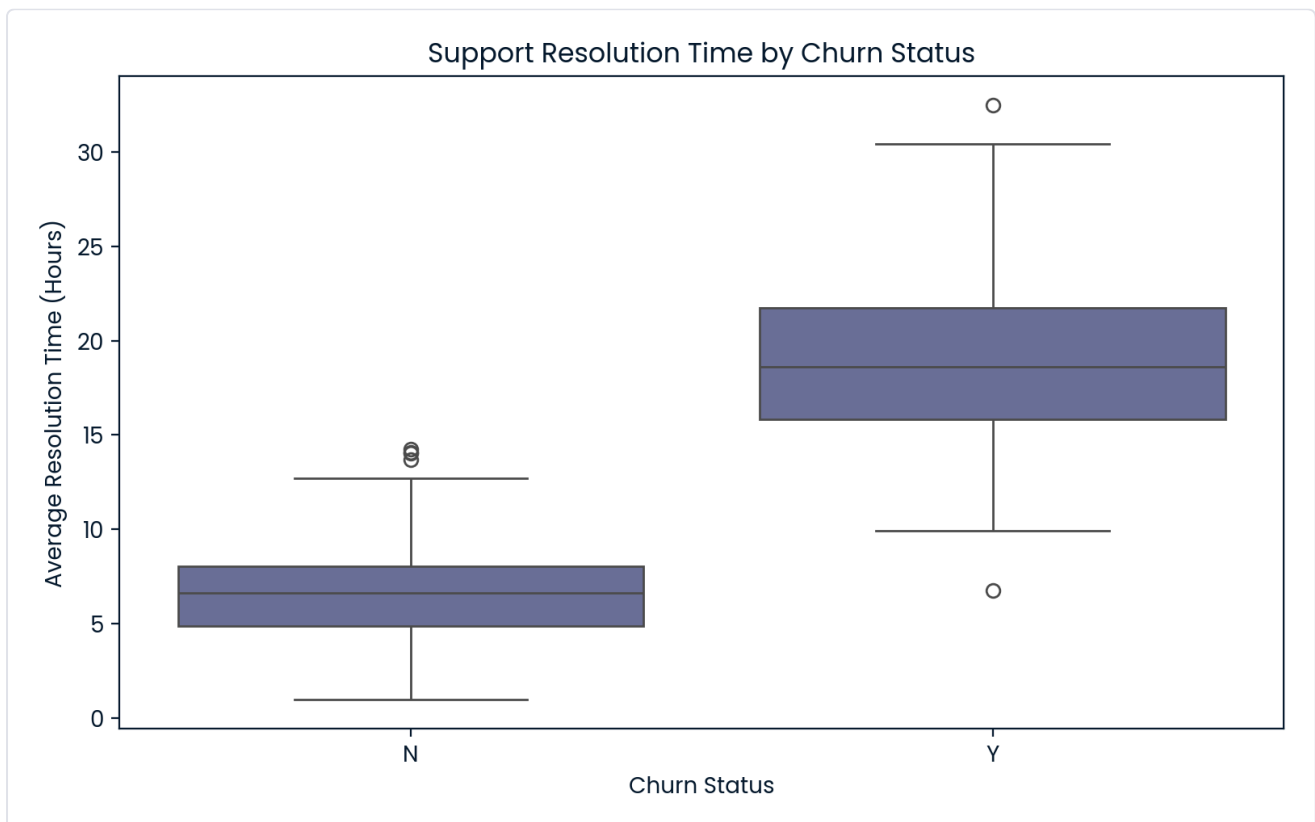
Important: the unit of analysis is the customer, rather than each individual support ticket. 33 customers without resolution time data were excluded from the analysis.

The average customer-level resolution time was 18.69 hours for the Churn group ($n = 109$), compared with 6.66 hours for the No Churn group ($n = 258$), representing a difference of approximately 12.03 hours.

Welch's t-test: $p < 0.001$; 95% confidence interval for the mean difference: approximately 11.09–12.96 hours. The data indicate a strong association between resolution time and churn. However, since subscription cancellation dates are unavailable, neither the temporal sequence nor causality can be established.

Nevertheless, the findings suggest that support ticket resolution time may be an important factor associated with customer churn and warrants further investigation.

Churn Status	Count	Mean	Median	Std
No Churn (N)	258	6.66	6.61	2.69
Churn (Y)	109	18.69	18.63	4.61



Number of Recorded Activities vs. Churn

Does the proportion of customers classified as churned differ across groups with other number recorded activity events?

- H_0 : Activity group and churn status are independent.
- H_1 : There is an association between activity group and churn status.

Important: The less frequent categories of 3, 4, and 5 recorded events were combined into a single 3+ category to avoid small expected frequencies in the chi-square (χ^2) test.

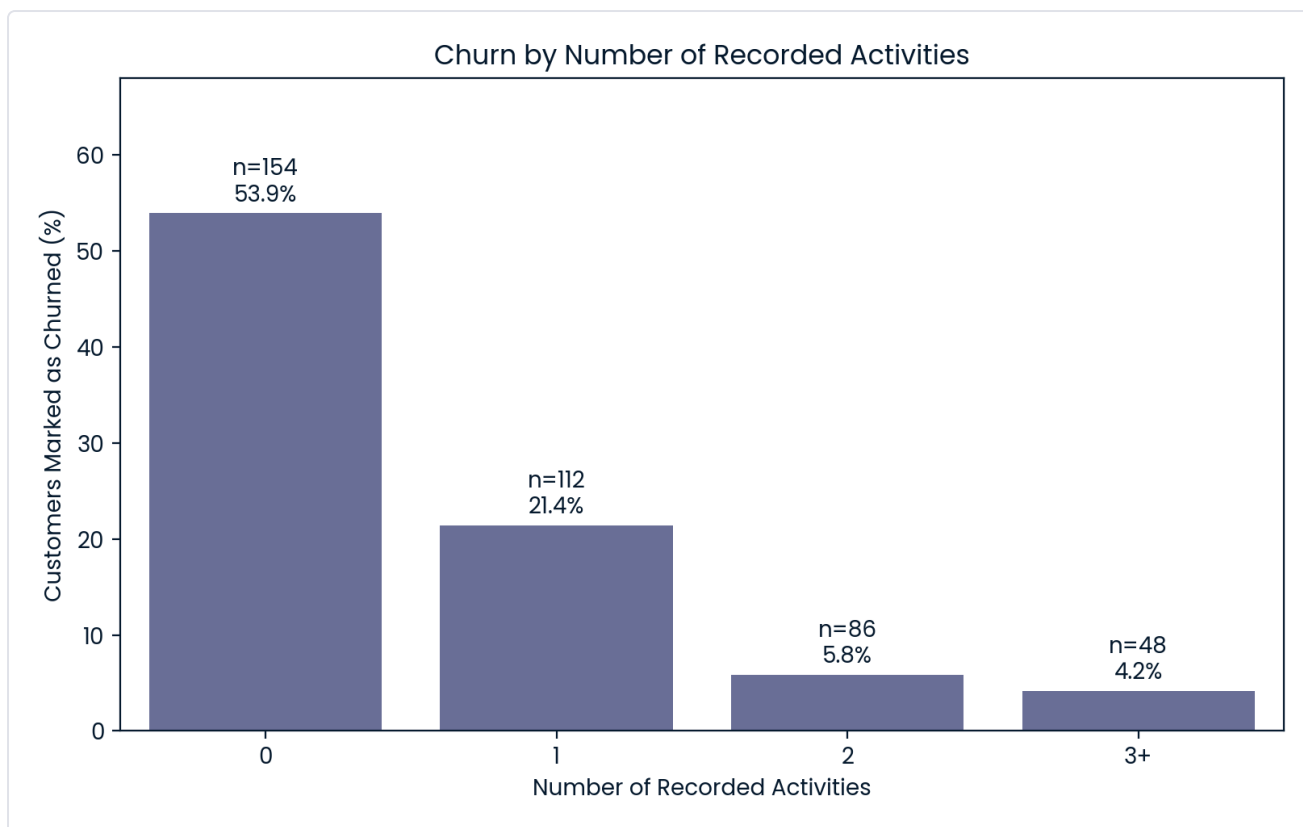
The proportion of customers classified as churned was 53.9% among those with 0 recorded activity events ($n = 154$), 21.4% with 1 event ($n = 112$), 5.8% with 2 events ($n = 86$), and 4.2% with 3+ events ($n = 48$).

The chi-square test ($\chi^2(3) = 87.16, p < 0.001$) indicates a statistically significant association between the number of recorded activities and churn status.

The absence of recorded activities refers to a lack of recorded events, which may have resulted from system downtime, rather than confirmed non-use of the application.

The findings suggest that customers with fewer recorded activity events are more frequently classified as churned.

Activity Group	No Churn (N)	Churn (Y)
0	71	83
1	88	24
2	81	5
3+	46	2



Activity Type vs. Churn

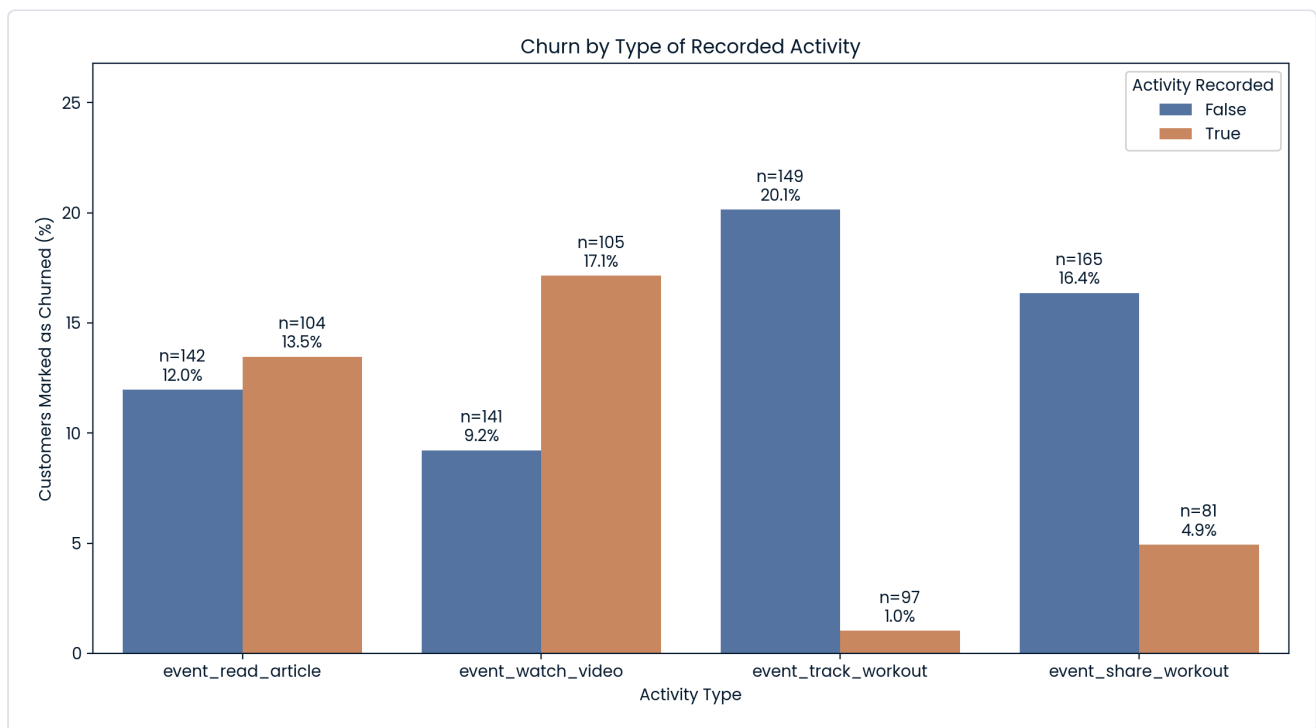
Is performing a specific type of activity associated with churn status among customers with at least one recorded activity event?

The comparisons include 246 customers with at least one recorded activity event. For each activity type, customers who performed that activity at least once (≥ 1 event) were compared with customers who performed other types of activities. The categories are not mutually exclusive.

Among customers with at least one recorded activity event, the churn rate was 1.0% for those who tracked workouts (`track_workout`), compared with 20.1% for other customers. For workout sharing (`share_workout`), the corresponding rates were 4.9% and 16.4%. After Holm's correction, both associations remained statistically significant ($p \approx 0.0000075$ and $p \approx 0.039$, respectively). No statistically significant associations were found for reading articles or watching videos after correction.

This does not prove that workout-related features themselves reduce churn, as the observed differences may reflect overall user engagement. However, the findings may suggest which types of activities could be promoted as part of customer retention strategies.

Activity Type	Has Event	Count (n)	Churn (%)
read_article	False	142	12.0
read_article	True	104	13.5
share_workout	False	165	16.4
share_workout	True	81	4.9
track_workout	False	149	20.1
track_workout	True	97	1.0
watch_video	False	141	9.2
watch_video	True	105	17.1



6. Business Metric and Baseline

Business Objective: Monitor customer cancellations and evaluate changes in retention.

Target KPI: Monthly Customer Churn Rate, defined as the number of customers who canceled their subscriptions during a given month, divided by the number of active customers at the beginning of that month, multiplied by 100%.

The provided dataset does not include cancellation dates or monthly active customer counts, making it impossible to calculate the monthly churn rate. The available data only provide a baseline measure that is not monthly: the proportion of customers classified as churned in the dataset.

It is important to note that this measure is not directly comparable to the future monthly churn rate.

Baseline value of the currently observable metric: $114 / 400 = 28.5\%$.

To implement the target KPI, cancellation dates and the number of active accounts at the beginning of each month must be collected.

Additional metrics to monitor:

- Percentage of customers with recorded activity within a fixed observation window.
- Average support ticket resolution time.

In the current dataset, these values are 61.5% of accounts with at least one recorded activity event during the available observation period and 10.39 hours per support ticket, respectively. These cross-sectional measures should not be directly interpreted as comprehensive monthly performance indicators.

7. Business Recommendations

The proposed actions are based on the findings of the analysis but require further testing. **Their impact on churn should not be assumed to be consistent, definitive, or causal.**

1. Improve Customer Support Efficiency

Customers classified as `Churn` had longer average support ticket resolution times than those classified as `No Churn`. However, simply having a support ticket was not significantly associated with churn. Therefore, the recommendation primarily concerns **the quality and duration of the support process**, rather than reducing the number of customer interactions.

Efforts should focus on reducing support ticket resolution times. Progress should be monitored by tracking average resolution times and churn rates across customer cohorts.

2. Test Initiatives to Encourage Regular App Usage, Particularly Workout-Related Features

The associations between activity levels and churn were statistically significant. Among customers with recorded activities, those who used `track_workout` and `share_workout` had lower churn rates.

It would be worth introducing initiatives that encourage customers to use these features, such as simplifying workout tracking or implementing reminders.

The effectiveness of these initiatives should be monitored through activity metrics measured over consistent observation periods and churn rates across customer cohorts.

However, **it should not be assumed that increased activity itself causes lower churn.**

3. Monitor Retention Separately by Subscription Plan and Investigate the Specific Characteristics of the Free Plan

In the analyzed dataset, the Free plan had the highest proportion of customers classified as `Churn`.

Before making changes to the subscription offering, it is necessary to confirm how cancellations are recorded for Free plan users.

It may also be worth introducing additional features or benefits that encourage customers to remain with the service.

The effectiveness of these initiatives should be monitored by tracking churn rates across customer cohorts and subscription plans.

4. Improve the Quality and Completeness of Event Tracking

Collect subscription cancellation dates and the number of active customers at the beginning of each month.

Monitor system downtime and potential gaps in activity logs, and regularly validate the quality of customer support data.

These improvements are necessary to enable reliable churn measurement and more accurate evaluation of customer retention strategies.